JEE Mains 2019 Chapter wise Question Bank

Mathematical Reasoning – Questions

Q1

If the Boolean expression

$(p \oplus q) \wedge (\sim p \odot q)$ is equivalent to

$p \wedge q$, where $\oplus, \odot \in \{\wedge, \vee\}$ then the

ordered pair $(\oplus, \odot)$ is:

(1) $(\vee, \wedge)$ (2) $(\vee, \vee)$

(3) $(\wedge, \vee)$ (4) $(\wedge, \wedge)$

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Q2

The logical statement

$[\sim (\sim p \vee q) \vee (p \wedge r)] \wedge (\sim p \wedge r)$

is equivalent to:

(1) $(\sim p \wedge \sim q) \wedge r$ (2) $\sim p \vee r$

(3) $(p \wedge r) \wedge \sim q$ (4) $(p \wedge \sim q) \vee r$

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Q3

Consider the statement: "P(n) : $n^2 - n + 41$ is prime." Then which one of the following is true?

(1) Both P(3) and P(5) are true.

(2) P(3) is false but P(5) is true.

(3) Both P(3) and P(5) are false.

(4) P(5) is false but P(3) is true.

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Q4

Consider the following three statements:

P : 5 is a prime number.

Q : 7 is a factor of 192.

R : L.C.M. of 5 and 7 is 35.

Then the truth value of which one of the following statements is true?

(1) $(\sim P) \vee (Q \wedge R)$ (2) $(P \wedge Q) \vee (\sim R)$

(3) $(\sim P) \wedge (\sim Q \wedge R)$ (4) $P \vee (\sim Q \wedge R)$

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Q5

If q is false and $p \wedge q \leftrightarrow r$ is true, then which one of the following statements is a tautology?

(1) $(p \vee r) \rightarrow (p \wedge r)$ (2) $(p \wedge r) \rightarrow (p \vee r)$

(3) $p \wedge r$ (4) $p \vee r$

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Q6

Contrapositive of the statement "If two numbers are not equal, then their squares are not equal". is :

(1) If the squares of two numbers are not equal, then the numbers are equal.

(2) If the squares of two numbers are equal, then the numbers are not equal.

(3) If the squares of two numbers are equal, then the numbers are equal.

(4) If the squares of two numbers are not equal, then the numbers are not equal.

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Q7

The Boolean expression

$\left((p \wedge q) \vee (p \vee \sim q)\right) \wedge (\sim p \wedge \sim q)$ is equivalent to :

(1) $p \wedge q$ (2) $p \wedge (\sim q)$

(3) $(\sim p) \wedge (\sim q)$ (4) $p \vee (\sim q)$

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Q8

The expression $\sim(\sim p \rightarrow q)$ is logically equivalent to :

(1) $\sim p \wedge \sim q$ (2) $p \wedge \sim q$

(3) $\sim p \wedge q$ (4) $p \wedge q$

12 Jan Evening

Q9

The contrapositive of the statement "If you are born in India, then you are a citizen of India", is :

(1) If you are not a citizen of India, then you are not born in India.

(2) If you are a citizen of India, then you are born in India.

(3) If you are born in India, then you are not a citizen of India.

(4) If you are not born in India, then you are not a citizen of India.

8 April Morning

Q10

If a point R(4, y, z) lies on the line segment joining the points P(2, –3, 4) and Q(8, 0, 10), then distance of R from the origin is :

(1) $2\sqrt{14}$ (2) $2\sqrt{21}$ (3) 6 (4) $\sqrt{53}$

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Q11

For any two statements p and q, the negation of the expression $p \vee (\sim p \wedge q)$ is:

(1) $\sim p \wedge \sim q$ (2) $p \wedge q$

(3) $p \leftrightarrow q$ (4) $\sim p \vee \sim q$

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Q12

If $p \Rightarrow (q \vee r)$ is false, then the truth values of p, q, r are respectively:

(1) F, T, T (2) T, F, F (3) T, T, F (4) F, F, F

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Q13

Which one of the following Boolean expressions is a tautology ?

(1) $(p \wedge q) \vee (p \wedge \sim q)$ (2) $(p \vee q) \vee (p \vee \sim q)$

(3) $(p \vee q) \wedge (p \vee \sim q)$ (4) $(p \vee q) \wedge (\sim p \vee \sim q)$

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Q14

The negation of the Boolean expression $\sim s \vee (\sim r \wedge s)$ is equivalent to :

(1) $\sim s \wedge \sim r$ (2) r (3) $s \vee r$ (4) $s \wedge r$

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Q15

If the truth value of the statement $p \rightarrow (\sim q \vee r)$ is false (F), then the truth values of the statements p, q, r are respectively.

(1) T, T, F (2) T, F, F

(3) T, F, T (4) F, T, T

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Q16

The Boolean expression $\sim(p \Rightarrow (\sim q))$ is equivalent to :

(1) $p \wedge q$ (2) $q \Rightarrow \sim p$

(3) $p \vee q$ (4) $(\sim p) \Rightarrow q$

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Mathematical Reasoning - Solutions

Q1

(3) Check each option

(1) $(p \vee q) \wedge (\sim p \wedge q) = (\sim p \wedge q)$

(2) $(p \vee q) \wedge (\sim p \vee q) = q$

(3) $(p \wedge q) \wedge (\sim p \vee q) = p \wedge q$

(4) $(p \wedge q) \wedge (\sim p \wedge q) = F$

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Q2

(3) Logical statement,

$$[\sim(\sim p \vee q) \vee (p \wedge r)] \wedge (\sim q \wedge r)$$

$$= [(p \wedge \sim q) \vee (p \wedge r)] \wedge (\sim q \wedge r)$$

$$= [(p \wedge \sim q) \wedge (\sim q \wedge r)] \vee [(p \wedge r) \wedge (\sim q \wedge r)]$$

$$= [p \wedge \sim q \wedge r] \vee [p \wedge r \wedge \sim q]$$

$$= (p \wedge \sim q) \wedge r$$

$$= (p \wedge r) \wedge \sim q$$

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Q3

(1) $P(n) = n^2 - n + 41$

$\Rightarrow$ $P(3) = 9 - 3 + 41 = 47$ (prime)

& $P(5) = 25 - 5 + 41 = 61$ (prime)

$\therefore$ P(c) and $P(5)$ are both prime i.e., true.

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Q4

(4) P is True, Q is False and R is True

(1) $(\sim P) \vee (Q \wedge R) \equiv F \vee (F \wedge T) \equiv F \vee F = F$

(2) $(P \wedge Q) \vee (\sim R) \equiv (T \wedge F) \vee (F) \equiv F \vee F = F$

(3) $(\sim P) \wedge (\sim Q \wedge R) \equiv F \wedge (T \wedge T) \equiv F \wedge T = F$

(4) $P \vee (\sim Q \wedge R) \equiv T \vee (T \wedge T) \equiv T \vee T = T$

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Q5

(2) q is false and $[(p \wedge q) \leftrightarrow r]$ is true

As $(p \wedge q)$ is false

$[\text{False} \leftrightarrow r]$ is true

Hence r is false

Option (1): says $p \vee r$,

Since r is false

Hence $(p \vee r)$ can either be true or false

Option (2): says $(p \wedge r) \rightarrow (p \vee r)$

$(p \wedge r)$ is false

Since, $F \rightarrow T$ is true and

$F \rightarrow F$ is also true

Hence, it is a tautology

Option (3): $(p \vee r) \rightarrow (p \wedge r)$

i.e. $(p \vee r) \rightarrow F$

It can either be true or false

Option (4): $(p \wedge r)$,

Since, r is false

Hence, $(p \wedge r)$ is false.

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Q6

(3) Contrapositive of "If A then B" is "If ~B then ~A". Hence contrapositive of "If two numbers are not equal, then their squares are not equal" is "If squares of two numbers are equal, then the two numbers are equal".

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Q7

(3) Consider the Boolean expression

$$((p \wedge q) \vee (p \vee \sim q)) \wedge (\sim p \wedge \sim q)$$

$$\equiv (p \vee \sim q) \wedge (\sim p \wedge \sim q)$$

$$\equiv ((p \vee \sim q) \wedge \sim p) \wedge ((p \vee \sim q) \wedge \sim q)$$

$$\equiv ((p \wedge \sim p) \vee (\sim q \wedge \sim p)) \wedge \sim q$$

$$\equiv (\sim p \wedge \sim q) \wedge \sim q \equiv (\sim p \wedge \sim q)$$

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Q8

(1) $\sim(\sim p \rightarrow Q) \equiv \sim(p \vee q) \equiv \sim p \wedge \sim q$

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Q9

(1) S: "If you are born in India, then you are a citizen of India."
Contrapositive of $p \rightarrow q$ is $\sim q \rightarrow \sim p$
So contrapositive of statement S will be :
"If you are not a citizen of India, then you are not born in India."

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Q10

(1) Here, P, Q, R are collinear

$\therefore\ \overrightarrow{PR} = \lambda \overrightarrow{PQ}$

$2\hat{i} + (y+3)\hat{j} + (z-4)\hat{k} = \lambda[6\hat{i} + 3\hat{j} + 6\hat{k}]$

$\Rightarrow 6\lambda = 2,\ y + 3 = 3\lambda,\ z - 4 = 6\lambda$

$\Rightarrow \lambda = \dfrac{1}{3}, y = -2, z = 6$

$\therefore$ Point R (4, – 2, 6)

Now, OR $= \sqrt{(4)^2 + (-2)^2 + (6)^2} = \sqrt{56} = 2\sqrt{14}$

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Q11

(1) $\sim(p \vee (\sim p \wedge q)) = \sim(\sim p \wedge q) \wedge \sim p$

$= (\sim q \vee p) \wedge \sim p$

$= \sim p \wedge (p \vee \sim q)$

$= (\sim q \wedge \sim p) \vee (p \wedge \sim p)$

$= (\sim p \wedge \sim q)$

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Q12

(2) For $p \Rightarrow q \vee r$ to be F.
r should be F and $p \Rightarrow q$ should be F
for $p \Rightarrow q$ to be F, $p \Rightarrow$ T and $q \Rightarrow$ F
$p, q, r \equiv$ T, F, F

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Q13

(2) $(p \vee q) \vee (p \vee \sim q) = p \vee (q \vee p) \vee \sim q$

$= (p \vee p) \vee (q \vee \sim q) = p \vee T = T$
Hence first statement is tautology.

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Q14

(4) $\sim s \vee (\sim r \wedge s) \equiv (\sim s \vee \sim r) \wedge (\sim s \vee s)$
$\equiv (\sim s \vee \sim r)$ $\qquad (\because \sim s \vee s)$ is tautology)
$\equiv \sim (s \wedge r)$
Hence, its negation is $s \wedge r$.

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Q15

(1) Given statement $p \rightarrow (\sim q \vee r)$ is False.
$\Rightarrow$ p is True and $\sim q \vee r$ is False
$\Rightarrow$ p is True and $\sim q$ is False and r is False
$\therefore$ truth values of p, q r are T, T, F respectively.

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Q16

(1) Given Boolean expression is,

$\sim(p \Rightarrow (\sim q))$ $\qquad \{\because\ p \Rightarrow q$ is same as $\sim p \vee q\}$

$\equiv \sim((\sim p) \vee (\sim q)) \equiv p \wedge q$

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